

Java2: Lesson 10 – Interview

1. Add **summary** at the top, **documentation** at the bottom and **comments** where necessary.
- Recordings and interviews are an exception, but they should be included in your submission code.
2. **Cite** the source of your information.
3. Refrain from including anything **you don't fully understand**.

1	100 points 5-min challenge: you will get full credit after you reproduce your work during a class or interview. Prompt the user to enter two numbers, then calculate and print out their multiplication. 1. Create a generic class that can handle arithmetic multiplication with different data types: int, double 2. Write a test code to detect the input data type of the entered numbers and produce their multiplication. *For more information, please visit https://www.youtube.com/watch?v=jUcAyZ5OUm0 <i>*Do not edit your recordings. The act of editing is considered a form of cheating.</i> <table border="1"> <tr> <td data-bbox="154 709 300 1136">Sample output</td><td data-bbox="300 709 1549 1136"> <pre>num1 (int or double): 7 num2 (int or double): 5 7 x 5 = 35 ----- num1 (int or double): 7.0 num2 (int or double): 5.0 7.0 x 5.0 =35.0 ----- num1 (int or double): a num2 (int or double): 5 done!</pre> </td></tr> <tr> <td data-bbox="154 1136 300 1251">File(s) to submit</td><td data-bbox="300 1136 1549 1251"> <pre>J210_1.java J210_1.png YouTube link</pre> </td></tr> </table>	Sample output	<pre>num1 (int or double): 7 num2 (int or double): 5 7 x 5 = 35 ----- num1 (int or double): 7.0 num2 (int or double): 5.0 7.0 x 5.0 =35.0 ----- num1 (int or double): a num2 (int or double): 5 done!</pre>	File(s) to submit	<pre>J210_1.java J210_1.png YouTube link</pre>
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import java.util.Scanner;

// Generic class for multiplication
class Multiplier<T extends Number> {
    private T num1;
    private T num2;

    // Constructor
    public Multiplier(T num1, T num2) {
        this.num1 = num1;
        this.num2 = num2;
    }

    // Method to multiply the two numbers
    public double multiply() {
        return num1.doubleValue() * num2.doubleValue();
    }
}

// Main class to demonstrate functionality
public class GenericMultiplication {
    public static void main(String[] args) {
        Scanner scanner = new Scanner(System.in);

        // Prompt the user for input
        System.out.print("Enter first number (int or double): ");
        String input1 = scanner.next();
        System.out.print("Enter second number (int or double): ");
        String input2 = scanner.next();

        try {
            // Try to parse inputs as integers
            if (input1.contains(".") || input2.contains(".")) {
                double num1 = Double.parseDouble(input1);
                double num2 = Double.parseDouble(input2);

                Multiplier<Double> multiplier = new Multiplier<>(num1, num2);
                System.out.printf("%s x %s = %.2f%n", num1, num2, multiplier.multiply());
            } else {
                int num1 = Integer.parseInt(input1);
                int num2 = Integer.parseInt(input2);

                Multiplier<Integer> multiplier = new Multiplier<>(num1, num2);
                System.out.printf("%d x %d = %d%n", num1, num2, (int) multiplier.multiply());
            }
        } catch (NumberFormatException e) {
            System.out.println("Invalid input. Please enter valid numbers.");
        }
    }
}

```

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        scanner.close();  
    }  
}
```